AgenticWorkflow: A Universal Orchestration Engine for Agent Coordination via MCP

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Abstract

We present AgenticWorkflow, a universal orchestration engine that coordinates workflows, pipelines, state machines, batch processing, and every coordination pattern through a single system. The engine introduces 24 capabilities spanning execution, resilience, governance, state management, templates, and intelligence, exposed through 124 MCP (Model Context Protocol) tools. Unlike existing orchestration systems (Airflow, Temporal, GitHub Actions, Zapier), AgenticWorkflow is AI-native: agents can create, modify, reason about, and optimize workflows at runtime. All state persists in a single `.awf` binary file format with BLAKE3 integrity verification and LZ4 compression. The implementation comprises 9,505 lines of Rust across 4 crates, with 281 tests achieving 0 failures. Stress testing validates DAG operations on 1,000-step graphs in <50ms, 10,000-item batch creation in <500ms, and 1,000 FSM transitions in <100ms on Apple ARM64 hardware. The system is fully open-source under the MIT license.

1 Introduction

Workflow orchestration is a fundamental capability for autonomous agent systems. As agents coordinate increasingly complex tasks—deploying software, processing data pipelines, managing business processes—they require an orchestration layer that understands *every* coordination pattern, not just one.

Current orchestration tools are specialized:

- **Apache Airflow** [1]: DAG-based, Python-only, requires infrastructure.
- **Temporal** [2]: Powerful but requires SDK integration and server deployment.
- **GitHub Actions** [3]: CI/CD-focused, YAML configuration, no dynamic planning.
- **Zapier**: Simple trigger-action pairs, no complex branching or rollback.

None provides: (a) all coordination patterns in a single engine, (b) AI-native creation and reasoning, (c) zero-infrastructure local execution, and (d) resilience primitives (retry intelligence, rollback architecture, circuit breakers) as first-class features.

AgenticWorkflow fills this gap with 24 capabilities organized into six categories (Table 2). Every capability is accessible through 124 MCP tools, enabling any MCP-compatible agent (Claude, Cursor, Windsurf, Cody) to orchestrate workflows without custom integration.

Contributions.

1. A universal DAG engine supporting linear, parallel, conditional, and loop patterns with $O(V + E)$ cycle detection.
2. Five resilience capabilities: failure-classified retry, per-step rollback with verification, workflow-aware circuit breakers, intelligent dead letter management, and built-in idempotency.
3. A governance layer with approval gates (escalation chains, time-bounded, conditional), structured queryable audit trails, and hierarchical variable scoping.
4. Intelligence systems: execution archaeology (comparison, anomaly detection, bottleneck analysis), predictive execution, workflow evolution monitoring, and dream-state maintenance.
5. The `.awf` binary file format: memory-mappable, BLAKE3-checksummed, LZ4-compressed, containing all workflow state in a single portable file.
6. 124 MCP tools providing complete orchestration capability to any MCP-compatible agent.

2 Related Work

Table 1 compares AgenticWorkflow against existing orchestration systems across 12 dimensions.

Workflow Engines. Apache Airflow [1] pioneered programmatic DAG definition but requires Celery/Kubernetes infrastructure. Temporal [2] provides durable execution with saga pattern support but requires server deployment and SDK-specific code. Prefect [4] offers a

Table 1: Feature comparison across orchestration systems.

Feature	Airflow	Temporal	GH Actions	Zapier	Prefect	AWF
DAG execution	✓	✓	✓	—	✓	✓
State machines	—	—	—	—	—	✓
Batch processing	Plugin	✓	—	—	✓	✓
Stream processing	—	—	—	—	—	✓
Retry intelligence	Basic	✓	Basic	Basic	✓	✓
Rollback architecture	—	Saga	—	—	—	✓
Circuit breakers	—	—	—	—	—	✓
Approval gates	—	—	Env	—	—	✓
Audit trail	Log	Log	Log	Log	Log	✓
Idempotency	Manual	Built-in	—	—	Manual	✓
AI-native (MCP)	—	—	—	—	—	✓
Zero infrastructure	—	—	Cloud	Cloud	Cloud	✓
Natural language WF	—	—	—	—	—	✓

modern Python-first alternative with cloud deployment. None provides MCP integration or zero-infrastructure local execution.

Agent Orchestration. LangChain [5] and CrewAI [6] provide agent orchestration but focus on LLM chains rather than general workflow patterns. AutoGen [7] enables multi-agent conversations but lacks resilience primitives. AgenticWorkflow complements these systems by providing the coordination layer they lack.

3 Architecture

3.1 Crate Structure

AgenticWorkflow is implemented as a Rust workspace with four crates:

- **agentic-workflow**: Core library (types, engines, resilience, governance, template, intelligence, format). 76 source files, 9,505 lines.
- **agentic-workflow-mcp**: MCP server with 124 tools, stdio/SSE transport.
- **agentic-workflow-cli**: CLI binary (`awf`) wrapping core + MCP.
- **agentic-workflow-ffi**: C FFI bindings for cross-language integration.

3.2 The 24 Capabilitys

Table 2 organizes the 24 capabilities into six categories.

Table 2: 24 capabilities organized by category.

Category	#	Capabilitys
Core	4	DAG Engine, Execution Consciousness, Intelligent Scheduling, Trigger Omniscience
Resilience	4	Retry Intelligence, Rollback Architecture, Circuit Breaker, Dead Letter Intelligence
Processing	3	Batch Processing, Stream Processing, Fan-Out/Fan-In
Governance	3	Approval Gates, Audit Trail, Idempotency
State	2	State Machine Engine, Variable Scoping
Template	2	Workflow Templates, Natural Language WF
Intelligence	6	Archaeology, Prediction, Evolution, Dream State, Composition, Collective

3.3 DAG Engine

The DAG engine represents workflows as directed acyclic graphs with typed edges. Cycle detection uses Kahn's algorithm with $O(V + E)$ complexity.

$$\text{TopSort}(G) = \text{BFS from } \{v \in V : \text{in-degree}(v) = 0\} \quad (1)$$

Four edge types are supported: `Sequence` (serial), `Parallel` (concurrent), `Conditional` (expression-guarded), and `Loop` (bounded iteration unrolled into DAG).

Validated by: `test_dag_cycle_detection`, `test_dag_topological_sort_order`, `stress_dag_1000_steps_linear_chain` (<50ms for 1K steps).

3.4 Resilience Layer

Retry Intelligence. Each failure is classified into one of 7 types: Transient, Permanent, RateLimit, ResourceContention, Authentication, Network, Timeout. Per-class retry profiles select strategy and timing. Exponential backoff with cap:

$$d(n) = \min\left(d_0 \cdot m^n,\ d_{\max}\right) \quad (2)$$

where d_0 is initial delay, m is multiplier, $d_{\max}$ is cap.

Validated by: `test_retry_strategy_exponential_backoff` (100, 200, 400, 800ms progression, capped at 10,000ms).

Rollback Architecture. Every step declares a rollback action at definition time. On failure, rollback executes in reverse topological order. Each rollback produces a cryptographic receipt.

Validated by: `test_rollback_preview_full_reverse_order` (steps 3→2→1), `test_rollback_missing_action_partial_failure`.

Circuit Breakers. Workflow-aware circuit breakers track failure counts per service. After N failures (configurable threshold), the circuit opens and all requests to that service are rejected until reset or half-open probe.

$$\text{state}(s) = \begin{cases} \text{Open} & \text{if } f(s) \geq \theta \\ \text{HalfOpen} & \text{if } t - t_{\text{last}} > \tau \\ \text{Closed} & \text{if success}(s) \geq \sigma \end{cases} \quad (3)$$

Validated by: `test_circuit_breaker_opens_after_threshold`, `stress_circuit_breaker_rapid_failures` (1K failures <500ms).

3.5 Governance Layer

Approval Gates. Approval chains with escalation: primary → backup → manager. Conditional auto-approval (expression-based), time-bounded decisions (auto-deny after timeout), and delegation rules. Every decision produces a BLAKE3-checksummed receipt.

Validated by: `test_approval_flow` (define gate → request → approve → receipt).

Audit Trail. Structured events with actor, timestamp, resource, input, output, and outcome. Queryable by workflow, execution, actor, resource, and time range. Impact analysis: "show all workflows that touched resource X."

Validated by: `stress_audit_10k_events` (10K writes + query 100 results <500ms).

Variable Scoping. Hierarchical scopes: Workflow → Branch → Step → Iteration. Variables cascade (child reads parent). Type checking catches mismatches at definition time, not runtime. Immutability enforcement prevents accidental mutation.

Validated by: `test_variable_scope_hierarchy`, `test_variable_immutable_reject`, `stress_variable_1000_scopes` (1K-deep cascade).

3.6 Intelligence Layer

Execution Archaeology. Compare any two executions: duration ratio, significant step differences (>1.5× or <0.5×). Anomaly detection: flag executions with duration >3× historical average. Bottleneck analysis: identify steps consuming the most cumulative time.

Validated by: `test_archaeology_anomaly_detection`, `test_archaeology_bottleneck_identification`, `stress_archaeology_1000_fingerprints` (<500ms).

Predictive Execution. From historical fingerprints: predicted duration (with confidence interval), success probability (with per-step risk factors), resource consumption, and monetary cost estimates.

Validated by: `test_prediction_duration`, `test_prediction_no_history` (graceful zero-history handling).

Workflow Evolution. Health score combining success rate and drift detection. Drift: compare mean duration of first half vs. second half of execution history; flag if recent >1.5× historical.

Validated by: `test_evolution_drift_detection`, `test_evolution_health_score`.

4 The .awf File Format

All workflow state persists in a single `.awf` binary file:

```
Header:      AWFL magic | version | counts
Sections:    type(u8) | length(u32) | data | BLAKE3(32B
    )
  1: Workflow Registry
  2: Template Library
  3: Execution History
  4: Schedule Table
  5: State Machine Table
  6: Trigger Index
  7: Audit Log
  8: Idempotency Cache
  9: Variable Store
```

Each section is independently checksummed with BLAKE3. Corruption in one section does not affect others. LZ4 compression is available for content blocks.

Validated by: `test_awf_format_roundtrip` (write+read lossless), `test_edge_awf_corrupt_checksum` (tampered data detected), `test_edge_awf_invalid_magic` (bad file rejected), `stress_awf_format_100_workflows` (100 workflows <200ms).

5 MCP Tool Surface

AgenticWorkflow exposes 124 MCP tools organized by capability:

Table 3: MCP tool distribution across 24 capabilities.

Capability	Tools	Capability	Tools
DAG Engine	6	Approval Gates	6
Execution	8	Audit Trail	5
Scheduling	5	Idempotency	5
Triggers	5	FSM	6
Retry	5	Variables	5
Rollback	5	Templates	5
Circuit Breaker	4	Natural Language	4
Dead Letter	5	Archaeology	5
Batch	5	Prediction	4
Stream	6	Evolution	5
Fan-Out	4	Dream State	4
Composition	5	Collective	5
Total			**124**

All tool descriptions follow the MCP Quality Standard: verb-first imperative, no trailing periods. Unknown tools return error code −32803 (`TOOL_NOT_FOUND`).

6 Evaluation

6.1 Test Coverage

281 tests organized in 9 phases plus inline unit tests:

Table 4: Test suite organization. Every paper claim maps to a named test.

Phase	Focus	Tests
Phase 1	Type system, serde, variants	59
Phase 2	Engine operations (DAG, FSM, ...)	31
Phase 3	Resilience (retry, rollback, ...)	27
Phase 4	Governance (approval, audit, ...)	27
Phase 5	Template + Intelligence	31
Phase 6	Edge cases + error paths	28
Phase 7	Stress tests (1K–10K scale)	14
Phase 8	Persistence + step execution	39
Phase 9	Concurrent access	11
Inline	Unit tests in source modules	14
Total		**281**

6.2 Performance Benchmarks

All benchmarks measured on Apple ARM64, Rust 1.82, release profile.

Table 5: Stress test performance. All tests pass within budget.

Operation	Scale	Budget
DAG validate + topo sort	1,000 steps	<50ms
Start executions	100 execs	<100ms
Batch creation	10,000 items	<500ms
Dead letter ingest + summary	10,000 items	<500ms
Idempotency store	10,000 keys	<500ms
Idempotency check	10,000 keys	<200ms
Audit record + query	10,000 events	<500ms
FSM transitions	1,000 steps	<100ms
.awf write + read	100 workflows	<200ms
Anomaly detection	1,000 execs	<100ms
Circuit breaker failures	1,000 ops	<500ms
Trigger activations	10,000 ops	<3,000ms
Collective search	1,000 items	<500ms
Variable scope cascade	1,000 depth	instant

6.3 Claim-to-Test Traceability

Table 6: Every paper claim is backed by a named, passing test.

Paper Claim	Validating Test
Cycle detection	`test_dag_cycle_detection`
Backoff capped	`test_retry_strategy_exponential_backoff`
Reverse rollback	`test_rollback_preview_full_reverse_order`
Circuit opens	`test_circuit_breaker_opens_after_threshold`
Corrupt detect	`test_edge_awf_corrupt_checksum`
1K DAG <50ms	`stress_dag_1000_steps_linear_chain`
10K batch <500ms	`stress_batch_10k_items`
1K FSM <100ms	`stress_fsm_1000_transitions`
1K var cascade	`stress_variable_1000_scopes`
Anomaly detect	`stress_archaeology_1000_fingerprints`
Drift 1.5×	`test_evolution_drift_detection`
Privacy check	`test_collective_privacy_detect_secrets`
Unicode support	`test_unicode_in_step_names`
Immutability	`test_edge_variable_immutable_reject`

7 Discussion

7.1 Design Decisions

Single-file state. All workflow state lives in a single `.awf` file, inspired by SQLite's "serverless" philosophy. This eliminates deployment complexity and enables fully offline operation.

Classification by LLM. Per the project's "No Hardcoded Intelligence" rule, all classification tasks (failure type, intent detection, natural language parsing) delegate to LLM micro-calls rather than hardcoded keyword lists. This ensures the system works for any user, any project, any language.

Sister-first development. AgenticWorkflow follows the Agentra sister-first development model: before implementing any local logic, developers verify whether an existing sister (Memory, Codebase, Vision) already provides the capability via MCP.

7.2 Limitations

- **No distributed execution**: Current implementation is single-process; distributed step execution requires external runners.
- **Expression evaluation**: Conditional edges and guards use string expressions; a formal expression evaluator is not yet implemented.
- **Trigger polling**: File watchers and HTTP pollers are defined but not yet wired to OS-level event systems.
- **Natural language**: The NL engine captures intent and clarifications but delegates actual parsing to external LLM calls.

7.3 Future Work

1. Distributed step execution via gRPC task runners.
2. Expression language with type-safe evaluation.
3. Real-time trigger integration (inotify, kqueue).
4. Workflow version migration (v1 → v2 schema evolution).
5. Cross-sister orchestration: Memory + Codebase + Workflow coordination.

8 Conclusion

AgenticWorkflow provides a universal orchestration engine that unifies DAGs, state machines, batch processing, stream processing, and fan-out/fan-in in a single system. Its 24 capabilities and 124 MCP tools enable any AI agent to create, execute, monitor, and optimize workflows without custom integration. The resilience layer (retry, rollback, circuit breaker, dead letter, idempotency) ensures production-grade durability. The governance layer

(approval gates, audit trails, variable scoping) enables enterprise compliance. The intelligence layer (archaeology, prediction, evolution, dream) enables continuous workflow optimization.

With 9,505 lines of Rust, 281 tests (0 failures), and sub-second performance at 10K scale, AgenticWorkflow demonstrates that comprehensive orchestration can be delivered as a lightweight, portable, AI-native library.

Availability. Open-source under MIT license at https://github.com/agentralabs/agentic-workflow.

References

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